Name+Email ID

CSCI 5521: Machine Learning Fundamentals (Spring 2022)

Quiz 2 (Thurs, Feb 24))

Due on Gradescope at 02:00 PM, Friday, Feb 25

Instructions:

- This quiz has 3 questions, 30 points, on 2 page.
- Please **write your name & ID on this cover page**.

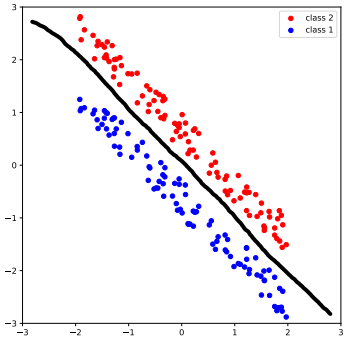
1. (**12 points**) For three data points $\begin{pmatrix}0\\10\end{pmatrix}, \begin{pmatrix}0\\5\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}$,

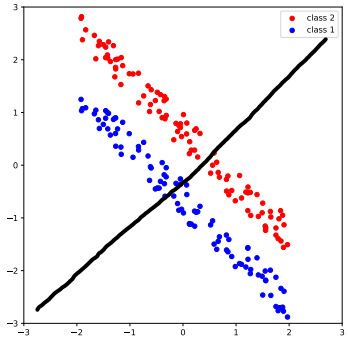
   (a) Derive the sample mean. 1/3 * (0 + 0 + 0, 10 + 5 + 0) = (0, 5)

   (b) Derive the **unbiased** sample covariance matrix.

   (c) Explain one of the diagonal entries in the covariance matrix (*e.g.*, if your $\sigma_{11} = c$, intuitively explain why it is equal to c here).

2. (**10 points**) In the following figures, (a) draw the first principal component direction in the left figure, and the first linear discriminant direction in the right figure. Briefly explain.





The PCA method centers and then sets the axis along the most variance in the data. The Linear discriminant chooses a direction that lets data of a class cluster closer together when projected onto the line.

(b) We are going to perform a binary classification on the data in the reduced 1-D space. Shall we project the data onto the direction found by PCA or LDA? Briefly explain.

LDA will clearly allow the two classes to become separately grouped.

3. (**8 points**) Select all the option(s) that are correct about K-means and EM for Gaussian Mixtures:

   (a) They are both clustering methods.

   (b) EM for Gaussian Mixtures has less parameters than K-means.

   (c) EM for Gaussian Mixtures is faster to compute than K-means.

   (d) K-means always finds local optimum while EM for Gaussian Mixtures always finds global optimum.